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Instruction

What the question is

My comments/Pay Attention

CNF, CFL Pumping Lemma & PDA

CNF Practice: 2.14, 2.26

2.14 Convert the following CFG into an equivalent CFG in Chomsky normal form, using the procedure given in Theorem 2.9.

$$\begin{aligned} A &\rightarrow BAB \mid B \mid \epsilon \\ B &\rightarrow 00 \mid \epsilon \end{aligned}$$

2.14] CNF - Chomsky Normal Form

$A \rightarrow BAB|B|E$

$B \rightarrow OOLC$

Construct new CFG

$A \rightarrow BC$

$A \rightarrow a$ a = terminal

$A, B \& C$ Variables

$S \rightarrow E$, the S = start variable

Let S 's start var S_0 & rule $S_0 \rightarrow A$

$S_0 \rightarrow A$

$A \rightarrow BAB|B|E$

$B \rightarrow OOLC$

New start var given that start var does NOT occur on the right-hand side

Remove all rules that contain E

Elim/Remove $A \rightarrow E$ & $B \rightarrow E$

$S_0 \rightarrow A|E$

$A \rightarrow BAB|BA|AB|A|B|BB$

$B \rightarrow OO$

$S_0 \rightarrow E$ is accepted. so is start var, CNF ← allowed

Remove unit

$A \rightarrow A$ & remove

$S_0 \rightarrow A|E$

$A \rightarrow BAB|BA|AB|B|BB$

$B \rightarrow OO$

remove $S \rightarrow B$

$S_0 \rightarrow A|E$

$A \rightarrow BAB|BA|AB|OO|BB$

$B \rightarrow OO$

H. Pt 2
 $\leftarrow$ Remove $S_0 \rightarrow S$
 $S_0 \rightarrow BAB|BA|AB|00|BB|E$
 $A \rightarrow BAB|BA|AB|00|BB$
 $B \rightarrow 00$
 replace terminal 0 by var U
 $S_0 \rightarrow BAB|BA|AB|UU|BB|E$
 $A \rightarrow BAB|BA|AB|UU|BB$
 $B \rightarrow UU$

~~CFG~~ CFG = Context free grammar
 CNF = Chomsky normal form

$U \rightarrow 0$
 shorten right hand side by rules w/ 2 var.
 shorten rules, replace $S_0 \rightarrow BAB$ w/ 2 rules $S_0 \rightarrow BA_1$ &
 $A_1 \rightarrow AB$ $A \rightarrow BAB$ rule replaced by 2 rule
 $S_0 \rightarrow BA_1$ & $A_1 \rightarrow AB$. Rule $A \rightarrow BAB$ replaced by 2 rule $A \rightarrow BA_2$
 & $A_2 \rightarrow AB$

Fin. CFG in CNF $= G = (V, \Sigma, R, S_0)$ start var = S_0
 set of var $V = \{S_0, S, B, U, A_1, A_2\}$
 set of term $\Sigma = \{0\}$, & the rules R are

$S_0 \rightarrow BA_1 | BA | SB | UU | BB | E$
 $A \rightarrow BA_1 | BA | SB | UU | BB$
 $B \rightarrow UU$

$U \rightarrow 0$
 $A_1 \rightarrow AB$

2.26 Show that if G is a CFG in Chomsky normal form, then for any string $w \in L(G)$ of length $n \geq 1$, exactly $2n - 1$ steps are required for any derivation of w .

2.26 | $G = CFG$ in CNF

$w \in L(G)$ $n \geq 1$ for str w - we need $(2n-1)$ steps for der. of str w , ^{precisely}

Let's do induction method

for $n=1$: str of length 1 in CNF. so $s \rightarrow a$ where $a \in \Sigma$ & s is start sym

of steps can be obtained $2n-1 = 2(1)-1 = 2-1 = 1$, Thus, true for $2n-1$ for $n=1$ steps are required to str w .

for $n=k$ - take str of length $k \geq n$ in CNF. so this will take $2k-1$ step

$$2n-1 = 2(k)-1 = 2k-1$$

So lang ≥ 1 term sym & has so length

$n=k+1$ in CNF

$n \geq 1$, Consider lang in CNF where deriv start sym (s)

$$s \rightarrow BC$$

$$B \rightarrow^* x$$

$$C \rightarrow^* y$$

length of str start w/ start sym s is $|w| = |xy|$ where $|x| \geq 0$ & $|y| \geq 0$
induct hypo: CNF the length of deriv

it has to be

$$1 + (2|x|-1) + (2|y|-1) = 2|x| + 2|y| + 1 - 1 - 1 = 2(|x| + |y|) - 1$$

here $n = |x| + |y|$

$B \rightarrow^* x$ has a L of $|x|$ & $C \rightarrow^* y$ has a L of $|y|$

Hence, it is proved that it req $2n-1$ steps for deriv of str $w \in L(G)$ CNF

Revised since it didn't come out correctly:

2.26] $G = CFG$ in CNF

$w \in L(G)$ $n \geq 1$ for str w - we need $(2n-1)$ steps for der. of w . ^{precisely}

let's do induction method

for $n=1$. str w length 1 in CNF. so $s \rightarrow a$ where $a \in \Sigma$ & s is start sym

of steps can be obtained $2n-1 = 2(1)-1 = 2-1 = 1$, Thus, true for $2n-1$ for $n=1$ steps are required to str w .

for $n=k$ ~~take~~ str w length $k \geq n$ in CNF. so this will take $2k-1$ step

$$2n-1 = 2(k)-1 = 2k-1$$

so lang ≥ 1 term sym & has so length

$n=k+1$ in CNF

$n > 1$, Consider lang in CNF where deriv start sym (s)

$$s \rightarrow BC$$

$$B \rightarrow^* x$$

$$C \rightarrow^* y$$

length of str start w/ start sym s is $|w| = |xy|$ where $|x| \geq 0$ & $|y| \geq 0$

induct hypo: CNF the length of deriv

it has to be

$$1 + (2|x|-1) + (2|y|-1) = 2|x| + 2|y| + 1 - 1 - 1 = 2(|x| + |y|) - 1$$

$$\text{here } n = |x| + |y|$$

$B \rightarrow^* x$ has a L of $|x|$ & $C \rightarrow^* y$ has a L of $|y|$

Hence, it is proved that it req $2n-1$ steps for deriv of str $w \in L(G)$ CNF

Pumping Lemma Practice: 2.30a, 2.31 (hint: use $s=0p12p0p$), 2.32

2.30 Use the pumping lemma to show that the following languages are not context free.

a. $\{0^n 1^n 0^n 1^n \mid n \geq 0\}$

A ~~language~~ ~~is not context free~~

A ~~language~~ ~~is not context free~~

a. ~~language~~ ~~is not context free~~

2.30a) $B = \{0^n 1^n 0^n \mid n \geq 0\}$

$L = \text{length}$ $\text{str} = \text{string}$

Let P be Pumping length of B given by P.Lemma

Show B is NOT CFL. It's enough to show that a str $s = 0^P 1^P 0^P$

can't be pumped. Consider s.s of form $UVXYZ$.

• If Both V & Y contain at most 1 type of alphabet sym, str will be in the form of UV^2XY^2Z . Rhs of $0's$ & $1's$, $u \neq L$. Thus str s can't be a member of B .

• If either V or Y contains more than 1 type of alphabet sym, str will be of the form uv^2xy^2z which doesn't have the sym in correct order. Hence the str, s cannot be a mem of B .

Since str s cannot be pumped w/out violating the Pumping lemma cond, B is not a CFL.

//After this this is the previous Questions that were omitted. These questions were level 1000 hard for me. Took me hours to figure out.

2.31 Let B be the language of all palindromes over $\{0,1\}$ containing equal numbers of 0s and 1s. Show that B is not context free.

2.31 Let B be lang for all palindrom over $\{0,1\}$ containing the same # of 0's & 1's. To prove B is not CFL by using contradiction.
Assume that B is a CFL. Pumping Lemma = P.L

Since B is a CFL, then by P.L, there's a p (the P. length) where, if z is any str in B of length at least p , then z may be / in 5 pieces of $z = uvxyz$ w/ these condit:

$$1. |vx| \geq 1 \quad 2. |vxy| \leq p \quad 3. i \geq 0, uv^i xy^i z \in A$$

(Not a part of z)

$$2. |vxy| \leq p$$

z is a mem of B of length at least p . $n=2$ str z Assume
 $z = 0^2 1^4 0^2$ $z = 00111100$ can be / into 5 by: val

$$\begin{array}{ccccc} 00 & 11 & 1 & 10 & 0 \\ u & v & x & y & z \end{array}$$

Assume the str

$$001111110100 = z'$$

obtained str z' is not a palindrome after applying 3rd cond of P.L $z' \notin B$.

Lang B not following cond 3 of P.L

Apply 3rd cond. for $i=2$

$$\begin{array}{ccccc} 00 & \left(\frac{11}{v}\right)^2 & 1 & \left(\frac{10}{y}\right)^2 & 0 \\ u & & x & & z \end{array}$$

$$\begin{array}{ccccc} 00 & \left(\frac{1111}{v}\right) & 1 & \left(\frac{1010}{y}\right) & 0 \\ u & & x & & z \end{array}$$

$$z = 0^n 1^{2n} 0^n$$

z is a mem of B of length at least p

Assume val of $n=2$ for str z . $z = 0^2 1^4 0^2$ $z = 00111100$

$$\begin{array}{ccccc} 0 & 0 & 11 & 1 & 00 \\ u & v & x & y & z \end{array}$$

3rd cond of P.L $i=2$

Assume str $0010111100 = z'$

z' Not Palindrome after apply 3rd cond of P.L &

$z' \notin B$. Pumped str. The # of 0's & 1's $\neq$. Thus, $B \notin CFL$

Thus, two tries clearly prove that B is not CFL.

2.32 Let $\Sigma = \{1, 2, 3, 4\}$ and $C = \{w \in \Sigma^* \mid \text{in } w, \text{ the number of 1s equals the number of 2s, and the number of 3s equals the number of 4s}\}$. Show that C is not context free.

2.32 | let's say the oppo. L is Context free. So, L has pumping length p .

$$S = 1^p 3^p 2^p 4^p \in L \text{ w/ } |S| \geq p$$

$uVXYZ$ exists such that

a) $uv^i xy^i z \in L$ for all $i \geq 0$

b) $|VY| > 0$

c) $VXY \leq p$

Prove by contradiction, no matter what val $uVXYZ$

C1: if VXY has a 1 = $uV^2XY^2z \notin L$. B/c it can't be the same # of 1's & 2's. hence c). VXY can't have any 2's.

C2: If VXY has a 2

VXY can't have any 1's

C3: If VXY has a 3

VXY can't have any 4's

C4: If VXY has a 4

VXY can't have any 3's

Thus, A, B equation are contradicting. you see L is not CFL.

PDA Practice: Build state diagram of PDA for each language described in 2.4.

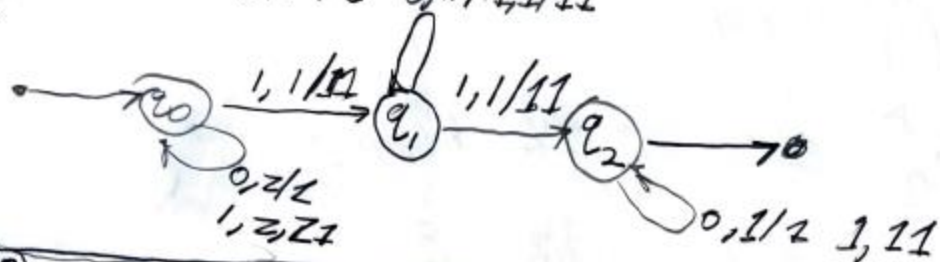
2.4 Give context-free grammars that generate the following languages. In all parts, the alphabet Σ is $\{0,1\}$.

- ^A**a.** $\{w \mid w \text{ contains at least three 1s}\}$
- b.** $\{w \mid w \text{ starts and ends with the same symbol}\}$
- c.** $\{w \mid \text{the length of } w \text{ is odd}\}$
- ^A**d.** $\{w \mid \text{the length of } w \text{ is odd and its middle symbol is a 0}\}$
- e.** $\{w \mid w = w^{\mathcal{R}}, \text{ that is, } w \text{ is a palindrome}\}$
- f.** The empty set

2.4 A)

$S \rightarrow X1X1X$

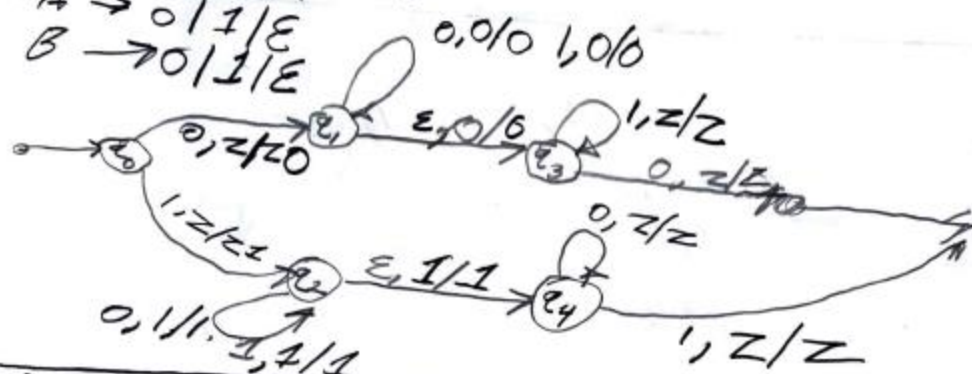
$X \rightarrow 0111 \epsilon$



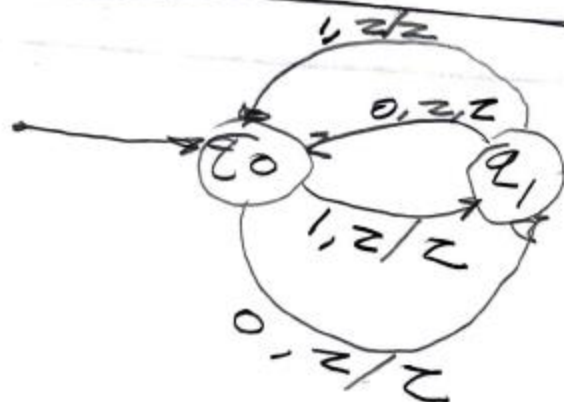
B) $S \rightarrow 0A011B1$

$A \rightarrow 011 \epsilon$

$B \rightarrow 011 \epsilon$



C)

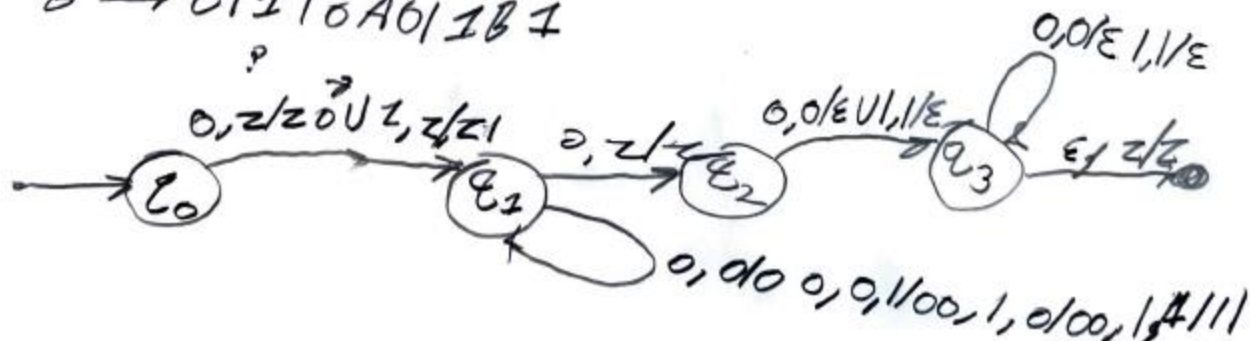


$S \rightarrow 01110A011B1$
 $A \rightarrow 01110A011B1$
 $B \rightarrow 01110A011B1$

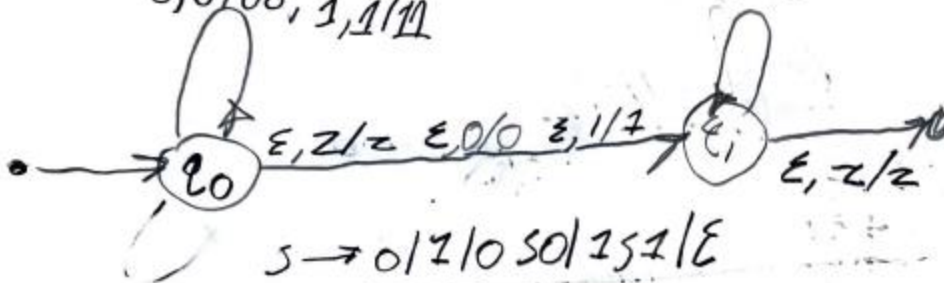
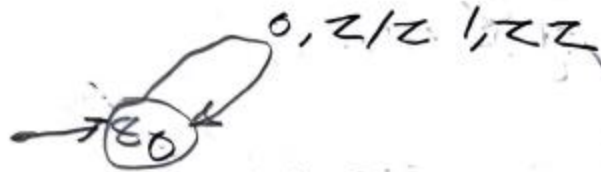
D) $S \rightarrow 01110A011B1$

$A \rightarrow 01110A011B1$

$B \rightarrow 01110A011B1$



0, 2/2, 0, 1/10
1, 2/11, 1, 0/01
0, 0/00, 1, 1/11

 $1, 2 \in$ 
$$\begin{aligned} S &\rightarrow X \\ X &\rightarrow YZ \\ Y &\rightarrow \epsilon \\ Z &\rightarrow 0|1 \end{aligned}$$


Build state diagram of PDA to recognize language of 2.9 (no big loop)

2.9 Give a context-free grammar that generates the language

$$A = \{a^i b^j c^k \mid i = j \text{ or } j = k \text{ where } i, j, k \geq 0\}.$$

Is your grammar ambiguous? Why or why not?

